

Kai Barker

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EDUCATION

University of California, Santa Barbara

Statistics and Data Science B.S.

Santa Barbara, CA

Graduating June 2026

Major GPA: 3.90/4.0

Activities: Consult Your Community, Statistics & Data Science Club, Data Science Collaborative, Ski & Snow Club

Coursework: Linear Algebra, Optimization, Machine Learning, Stochastic Processes, Time Series, Adv. Statistical Models, Big Data

SKILLS, CERTIFICATES & INTERESTS

Languages & Tools: Python (Pandas, Numpy, etc.), R, SQL, Java, C++, HTML, CSS, Tableau, Excel, Git, API Integration, Spark, LaTeX

Skills: Statistical Modeling, Time Series Forecasting, Feature Engineering, Machine Learning (Scikit, PyTorch), Pipeline Development

Certificates: JP Morgan Chase Software Engineering, JP Morgan Chase Quantitative Analysis, HackerRank SQL

Interests: Basketball, Hiking, Backpacking, Rock Climbing, Cooking, Running, Pickleball

EXPERIENCE

Engagement Manager

Consult Your Community

Santa Barbara, CA

February 2026 – Present

- Built a statewide financial model across all 115 CA community colleges using SCFF apportionment data, identifying that a 12% reduction in gateway math DFW rates generates \$26M in recoverable state funding.
- Designed a Tableau dashboard visualizing DFW rates, math success, and recoverable funding across CA community college market.
- Led a 16-week EdTech market entry engagement for an AI tutoring startup, directing a 7-person team across 3 research workstreams to deliver a market entry strategy and value proposition targeting the client's first paid B2B pilot.

Business Analyst

Consult Your Community

Santa Barbara, CA

October 2025 - Present

- Extracted and merged transactional data from Shopify, Square, and supplementary client records, normalizing and segmenting corporate accounts by key engagement metrics to identify revenue gaps and dormant clients.
- Translated findings into actionable recommendations including a CRM system, automated email reactivation campaigns, and a buyer-ready SOP, projected to recover at least \$40,000 in annual revenue.
- Built a Tableau dashboard with 10+ visualizations to communicate findings to a non-technical client and drive recommendations.

Quantitative Researcher

Department of Statistics and Applied Probability

Santa Barbara, CA

September 2024 – Present

- Analyzed 80,000+ congressional trade disclosures to identify statistically significant abnormal returns, validating a weighted alpha signal for portfolio construction.
- Applied Random Matrix Theory (Marchenko-Pastur) and Ledoit-Wolf shrinkage to clean a 560-asset covariance matrix, isolating 30 signal eigenvalues explaining 60.1% of variance.
- Optimized mean-variance portfolios in Python using cvxpy, achieving a 13.81% annual backtest return and +1.07% mean alpha (excluding 2022) vs. SPY over an 8-year walk-forward period.

Research Assistant

Foundations of Financial Technology Lab

Santa Barbara, CA

September 2024 – Present

- Analyzed 20M+ Ethereum blocks and 100M+ wallets to quantify wealth distribution, calculated a Gini coefficient of 0.96.
- Produced visualizations tracking inequality evolution over 5 years, revealing the top 1% of wallets control 95%+ of Ether supply.
- Filtered institutional and Sybil addresses to isolate organic wallet behavior and improve the accuracy of distribution metrics.
- Acknowledged in a published paper on blockchain wealth inequality with researchers from Yale, Cornell, and UCSB.

PROJECTS

NeurIPS Ariel Data Challenge

Competition - ESA Ariel Mission

- Built a signal processing pipeline (ADC conversion, sigma-clipped pixel masking, dark and flat field correction) to calibrate 200+ GB of raw telescope data across 14,000 files and 1,100+ exoplanets.
- Engineered a weighted ensemble of Gaussian Process Regression (composite RBF + Matern kernel), XGBoost, and Ridge to predict atmospheric spectra across 283 wavelength channels.
- Calibrated prediction uncertainty via GLL-optimal grid search over scale parameters, achieving a final score of 0.311.

MCMC & Simulated Annealing - Optimization Research

Academic Research

- Implemented Metropolis-Hastings MCMC in Python for message decryption, constructing the target likelihood from letter-to-letter transition probabilities trained on War and Peace text.
- Extended the algorithm with a simulated annealing modification (exponential cooling schedule, $\alpha=0.995$) over 50,000 iterations, reducing average incorrect letters from 22.3 to 21.6 vs. baseline.